

ECE-600-001

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Group: Jupiter

Checkpoint 6

This assignment focused on applying LiDAR-based sensing and real-world robot control to perform mapping and person-following tasks. We learned how LiDAR works to detect distances and generate a map of the environment, and how to connect and control the robot using the keyboard through ROS. By bringing up the robot and running SLAM (Simultaneous Localization and Mapping), we were able to map a live indoor environment in real time. This demonstrated how laser scans, odometry, and robot motion work together to build an occupancy grid map of the room.

In the second part of the assignment, we brought the robot up again and reconfigured its PID controller to safely follow a person. We adjusted PID parameters to ensure smooth and stable motion, preventing overshooting or unsafe movements. After tuning, we automated the person-following behavior so the robot could maintain a safe distance and adjust its speed and direction accordingly. Overall, this assignment reinforced key concepts in LiDAR sensing, ROS-based robot control, mapping, localization, and feedback control for safe autonomous behavior.

This checkpoint essentially has two parts to it

- a) Mapping
- b) Following a person

A) Mapping

- 1) In order to do this, we had to bring up the robot

```
roslaunch turtlebot3_gazebo turtlebot3_world.launch
```

- 2) launched the turtleslam turtlebot3_world.launch

```
roslaunch turtlebot3_slam turtlebot3_slam.launch
```

3 launched rViz

```
roslaunch turtlebot3_gazebo turtlebot3_gazebo_rviz.launch
```

4. we launched gazebo and then launched with teleop from the teleop keyboard to control the robot to map out the room

```
roslaunch turtlebot3_gazebo turtlebot3_gazebo_rviz.launch  
roslaunch turtlebot3_teleop turtlebot3_teleop_key.launch
```

That took a little while, because we had to become familiar with the layout of the keyboard, reminding ourselves it was not an RC car. But a forward/backward rotate left rotates right, and increases in speed, decrease in speed.

Figure 1: Using the keyboard to control the turtlebot

```
zaydhossain — jetson@yahboom: ~ — ssh jetson@10.250.88.232 — 80x24

J   K   L
M   <   >

t : up (+z)
b : down (-z)

anything else : stop

q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%

CTRL-C to quit

currently:      speed 0.308852563011      turn 0.51050010415
currently:      speed 0.308852563011      turn 0.561550114565
currently:      speed 0.308852563011      turn 0.617705126022
currently:      speed 0.308852563011      turn 0.679475638624
currently:      speed 0.308852563011      turn 0.747423202487
currently:      speed 0.308852563011      turn 0.822165522735
currently:      speed 0.308852563011      turn 0.904382075009
currently:      speed 0.339737819312      turn 0.904382075009
currently:      speed 0.339737819312      turn 0.99482028251
```

5 when the mapping was done

```
roslaunch map_server map_saver -f my_map
```

```
zaydhossain — jetson@yahboom: ~/yahboomcar_ws — ssh jetson@10.25...
Music
opencv_apps
Pictures
program
Public
Version.txt
Videos
voice_ctrl_test.py
yahboomcar_ws
yahboom.jpg

[jetson@yahboom:~$ cd yahboomcar_ws
[jetson@yahboom:~/yahboomcar_ws$ ls
build  devel  src  X3
[jetson@yahboom:~/yahboomcar_ws$ rosrn map_server map_saver -f
Usage:
  map_saver -h
  map_saver [--occ <threshold_occupied>] [--free <threshold_free>] [-f <mapname>
] [ROS remapping args]
[jetson@yahboom:~/yahboomcar_ws$ rosrn map_server map_saver -f ~/yahboomcar_ws/s
rc/yahboomcar_nav/maps/my_map
[ INFO] [1771275211.166363950]: Waiting for the map
[ INFO] [1771275211.456704210]: Received a 800 X 800 map @ 0.050 m/pix
[ INFO] [1771275211.456823535]: Writing map occupancy data to /home/jetson/yahbo
omcar_ws/src/yahboomcar_nav/maps/my_map.pgm
[ INFO] [1771275211.515668138]: Writing map occupancy data to /home/jetson/yahbo
omcar_ws/src/yahboomcar_nav/maps/my_map.yaml
[ INFO] [1771275211.516026685]: Done

jetson@yahboom:~/yahboomcar_ws$
```

Figure 2: After navigating the classroom while using gmap, the map is saved

```
zaydhossain — jetson@yahboom: ~/yahboomcar_ws/src/yahboomcar_na...
rc/yahboomcar_nav/maps/my_map
[ INFO] [1771275211.166363950]: Waiting for the map
[ INFO] [1771275211.456704210]: Received a 800 X 800 map @ 0.050 m/pix
[ INFO] [1771275211.456823535]: Writing map occupancy data to /home/jetson/yahbo
omcar_ws/src/yahboomcar_nav/maps/my_map.pgm
[ INFO] [1771275211.515668138]: Writing map occupancy data to /home/jetson/yahbo
omcar_ws/src/yahboomcar_nav/maps/my_map.yaml
[ INFO] [1771275211.516026685]: Done

[jetson@yahboom:~/yahboomcar_ws$ cd src/yahboomcar_nav/maps/my_map
-bash: cd: src/yahboomcar_nav/maps/my_map: No such file or directory
[jetson@yahboom:~/yahboomcar_ws$ ls
build  devel  src  X3
[jetson@yahboom:~/yahboomcar_ws$ cd src
[jetson@yahboom:~/yahboomcar_ws/src$ cd yahboomcar_nav
[jetson@yahboom:~/yahboomcar_ws/src/yahboomcar_nav$ cd maps
[jetson@yahboom:~/yahboomcar_ws/src/yahboomcar_nav/maps$ cd my_map
-bash: cd: my_map: No such file or directory
[jetson@yahboom:~/yahboomcar_ws/src/yahboomcar_nav/maps$ ls
carto_map.pbstream  house.pgm  map.yaml  my.yaml  testa.yaml
carto_map.pgm       house.yaml  my_map.pgm  rrt_map.pgm  testt.pgm
carto_map.sh        map.pgm    my_map.yaml  rrt_map.yaml  testt.yaml
carto_map.yaml      map.sh     my.pgm      testa.pgm
jetson@yahboom:~/yahboomcar_ws/src/yahboomcar_nav/maps$
```

Figure 3: Map saved as my_map.yaml

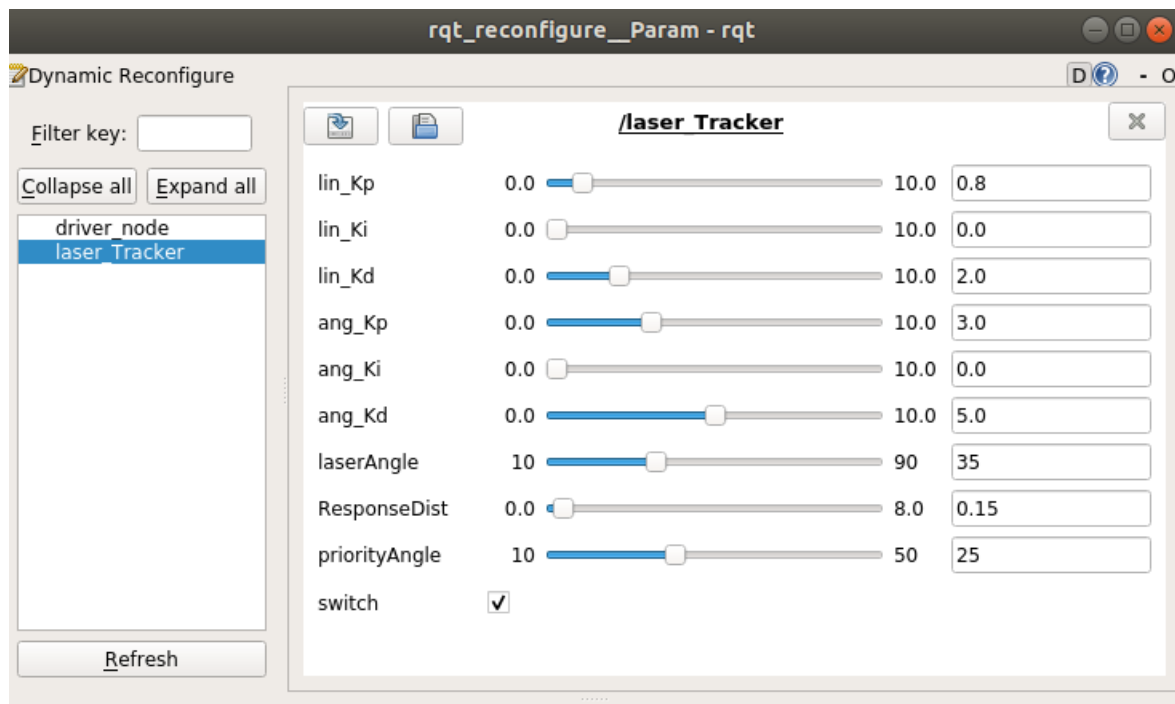
B) Follow a person

1) Robot bringup

```
roslaunch <robot_bringup>
```

2) Tuning the PID,

That took a while, a lot of trial and error, a lot of experimentation. The video of trial.mp4 was the PID in its final state BEFORE we adjusted the internal parameters to fit the “tested ones” as shown here



(System message might be shown here when necessary)

- 3) Loaded the tested parameters into the default parameters of the yaboomal robot for the autonomy of following a person. Video automation video.